MUHAMMAD ASAD FEROZI

Software Engineer

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RESEARCH INTERESTS

Applied machine learning and large language models; retrieval-augmented generation and grounded question answering; trustworthy and secure AI; computer vision for assistive technology; real-time and distributed systems for AI applications.

I am a software engineer at INTECH, building EmpowerGPT, an enterprise generative-AI platform: assistant builders, hybrid document search and multi-tenant sign-in. I hold a BS in Information Technology (CGPA 3.78/4.00) and am seeking a research-based master's to study how to make LLM systems grounded, measurable and safe to connect to real tools.

EDUCATION

Bachelor of Science in Information Technology 2021 – 2025

Punjab University College of Information Technology (PUCIT), University of the Punjab, Lahore **CGPA 3.78 / 4.00**

Relevant coursework: Artificial Intelligence, Machine Learning, Design and Analysis of Algorithms, Data Structures and Algorithms, Discrete Structures, Linear Algebra, Probability and Statistics, Calculus, Database Systems, Computer Networks, Operating Systems, Information Security, Cyber Security, Software Quality Assurance.

Final-year project: Parental Eye, a real-time tracking system with geofencing (see Projects).

PROFESSIONAL EXPERIENCE

Software Engineer, INTECH Automation and Intelligence, Lahore Jun 2026 – Present

- Led the AI Assistants revamp for **EmpowerGPT**: REST APIs for creating and sharing custom assistants, enterprise-knowledge and image-generation settings, and a redesigned assistant builder and knowledge-upload flow.
- Built Advanced Search and Explorer end to end (FastAPI, React): hybrid search across enterprise documents with suggestions, recent-activity tracking and workspace navigation, backed by indexing on multi-tenant PostgreSQL tables.

Associate Software Engineer, INTECH Automation and Intelligence, Lahore Jun 2025 – Jun 2026

- Led the Data Library revamp across three versions: file and folder browsing with permissions, uploads for CSV, audio and video with size limits and cancelling, and translations.
- Designed multi-tenant sign-in on Keycloak: domain-based tenant discovery, credential-reset rules for new tenants and Azure AD language sync for enterprise SSO.
- Implemented rate limiting for AI image generation across the chat and core backend services.
- Drove a configuration refactor across about 10 microservices and the frontend for multi-environment Kubernetes deployments, and contributed to the shared Python framework used by every backend service.
- Authored 145 pull requests (135 merged) on the platform since October 2025.

Full-stack Developer, Webox Systems, Lahore Jul 2024 – Jul 2025

- Developed **Euthyna**, a sustainability analytics platform that automates ESG gap analysis and compliance checks (PostgreSQL, Express, React, Node.js).
- Code-splitting, database indexing and API optimisation cut page load times by 30%; designed PostgreSQL schemas across 15+ tables, improving query efficiency by 30%.

PERN Stack Fellow, Webox Systems, Lahore · **Frontend Fellow**, Bytewise Limited (remote) 2023 – 2024

- Two fellowships (Mar–Jul 2023, Feb–May 2024) building full-stack and React applications in Agile teams.

PROJECTS AND RESEARCH

AI Lawyer: Retrieval-Augmented Legal Assistant, with a Retrieval Evaluation 2024; evaluation 2026

React · Express · LangChain · OpenAI embeddings · Pinecone · PostgreSQL · github.com/maferozi/AI-LAWYER-EXPRESS

- Built a legal question-answering assistant for Pakistani criminal law that answers from the statute rather than from the model's memory: the Pakistan Penal Code and a criminal-investigation handbook are split into chunks, embedded with OpenAI and indexed in Pinecone.
- Each question retrieves the ten closest passages and a LangChain QA chain answers from them; users sign in with JWT and their conversations are stored in PostgreSQL.
- Evaluated retrieval with 40 labelled everyday questions across 30 configurations (keyword, dense and hybrid search; chunks of 40–800 words). Dense retrieval with 200-word chunks found the answering section in the top 10 for 32 of 40 questions, against 25 of 40 at the chunk sizes the app used.
- Found that every model missed homicide questions because the Code uses terms such as *qatl-i-amd* and *qisas*, pointing to query expansion between everyday and legal vocabulary as the next step.

Parental Eye: Real-Time GPS Tracking with Geofencing · Final-year project, team of four Oct 2024 – Jun 2025

React · Leaflet · Node.js · MQTT · Socket.IO · PostgreSQL/PostGIS · Turf.js · parental-eye-react.vercel.app

- Built a system that shows parents and school transport operators where a child or vehicle is and alerts them when it leaves a safe area; wrote 108 of the team's 118 commits.
- Devices publish GPS points over MQTT; the server checks each point against circle, polygon and route geofences with Turf.js, stores it in PostGIS marked safe or danger, and pushes alerts over Socket.IO and email.
- Added a five-minute alert cooldown per user and device to stop alert floods, and marks a device inactive after 30 seconds of silence so stale positions are never shown as live.
- Built role-based dashboards for five roles (super admin, admin, guardian, driver, child) with live maps, geofence drawing on Leaflet, location history and printable reports.

Be My Eye: Obstacle Detection for the Visually Impaired Jan 2024 – Apr 2024

Python · Computer vision · Roboflow · Google Colab

- Trained a real-time obstacle-detection model to help visually impaired users move around safely, reaching 90% accuracy, 92% precision and 91% recall.

URL Shortener API · Job Board 2024 – 2025

- URL shortener REST API (Express 5, MongoDB, nanoid) with create, update, delete and access statistics, plus request validation.
- Laravel 11 job board with employers and tags, policy-based permissions and queued email notifications.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Java, C++, PHP
AI and Machine Learning	Retrieval-augmented generation, LangChain, OpenAI API, vector databases (Pinecone), hybrid search, computer vision
Back End	FastAPI, Node.js, Express, REST, WebSockets, MQTT, rate limiting, microservices
Data	PostgreSQL (partitioning, indexing), PostGIS, Redis, MongoDB
Front End	React, Redux Toolkit, Tailwind CSS, AG Grid, Leaflet
Platform	Kubernetes, Docker, Keycloak, Azure AD SSO, GitHub Actions, Nginx, Linux

CERTIFICATION

Supervised Machine Learning: Regression and Classification, Stanford University and DeepLearning.AI (Coursera), January 2025

LANGUAGES

English: professional working proficiency; all four years of the BS were taught in English (Medium of Instruction certificate available). **Urdu:** native.

REFERENCES

Available on request.